

# KE LI

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## EDUCATION

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| <b>Cornell University</b>                                                                                                                                                                                                                                                                                                                                                            | Jul. 2021 – Dec. 2026 (Expected) |
| <i>Ph.D. Candidate at <a href="#">SciFi Lab</a> in Information Science, College of Computing and Information Science</i>                                                                                                                                                                                                                                                             |                                  |
| <ul style="list-style-type: none"><li>• <b>Advisor:</b> Prof. <a href="#">Cheng Zhang</a>, <b>Committee Members:</b> Profs. <a href="#">Deborah Estrin</a> and <a href="#">Rajalakshmi Nandakumar</a></li><li>• <b>Concentration:</b> Ubiquitous Computing, Wearable Devices, Human-Computer Interaction, Artificial Intelligence</li><li>• <b>Minor:</b> Computer Science</li></ul> |                                  |
| <b>Cornell University</b>                                                                                                                                                                                                                                                                                                                                                            | Jul. 2021 - Dec. 2024            |
| <i>M.S. in Information Science, College of Computing and Information Science</i>                                                                                                                                                                                                                                                                                                     |                                  |
| <b>Shanghai Jiao Tong University (SJTU)</b>                                                                                                                                                                                                                                                                                                                                          | Sept. 2016 - Jul. 2020           |
| <i>B.S. in Information Engineering, School of Electronic Information and Electrical Engineering</i>                                                                                                                                                                                                                                                                                  |                                  |
| <ul style="list-style-type: none"><li>• <b>Overall GPA:</b> 91.41/100 (Rank 3/158)</li><li>• Outstanding Graduate from Universities in Shanghai (<b>Top 5%</b>)</li></ul>                                                                                                                                                                                                            |                                  |

## WORK EXPERIENCES

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| <b>Google</b>                                                                                                                  | May 2026 - Aug. 2026   |
| <i>Research Intern in the domain of multimodal sensing foundation research</i>                                                 |                        |
| <b>Meta</b>                                                                                                                    | Aug. 2025 - Dec. 2025  |
| <i>Research Intern in computer vision, machine learning, sensor fusion, and HCI software and hardware development</i>          |                        |
| <b>Carnegie Mellon University (CMU)</b>                                                                                        | Jul. 2019 - Sept. 2019 |
| <i>Research Assistant in Department of Electrical and Computer Engineering, Supervisor: Prof. <a href="#">Swarun Kumar</a></i> |                        |

## PUBLICATIONS

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| <a href="#">Head and Body Pose Estimation using Ultrasound Sensing on Smart Glasses</a> . <b>Ke Li</b> , Robin Kips, Filippo Arcadu, Marek Schikora, John Ho, Chris Graf, Jim Chen, Xingxing Cai, Gawsalyan Sivapalan, Ariyan Zarei, and Yan Deblangey. In <i>Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)</i> , 6 pages. | Barcelona, Spain<br>Apr. 2026 |
| <a href="#">µTouch: Enabling Accurate, Lightweight Self-Touch Sensing with Passive Magnets</a> . Siyuan Wang, <b>Ke Li</b> , Jingyuan Huang, Jike Wang, Cheng Zhang, Alanson P. Sample, and Dongyao Chen. In <i>2026 IEEE International Conference on Pervasive Computing and Communications (PerCom '26)</i> , pp. 1-10.                                                     | Pisa, Italy<br>Mar. 2026      |
| <a href="#">SonicID: User Identification on Smart Glasses with Acoustic Sensing</a> . <b>Ke Li</b> , Devansh Agarwal, Ruidong Zhang, Vipin Gunda, Tianjun Mo, Saif Mahmud, Boao Chen, François Guimbretière, and Cheng Zhang. <i>Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. (IMWUT '24)</i> 8, 4, Article 169, 27 pages.                                           | Nov. 2024                     |
| <a href="#">ActSonic: Recognizing Everyday Activities from Inaudible Acoustic Wave Around the Body</a> . Saif Mahmud, Vineet A Parikh, Qikang Liang, <b>Ke Li</b> , Ruidong Zhang, Ashwin Ajit, Vipin Gunda, Devansh Agarwal, François Guimbretière, and Cheng Zhang. <i>Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. (IMWUT '24)</i> 8, 4, Article 183, 32 pages.   | Nov. 2024                     |
| <a href="#">Ring-a-Pose: A Ring for Continuous Hand Pose Tracking</a> . Tianhong Catherine Yu, Guilin Hu, Ruidong Zhang, Hyunchul Lim, Saif Mahmud, Chi-Jung Lee, <b>Ke Li</b> , Devansh Agarwal, Shuyang Nie, Jinseok Oh, François Guimbretière, and Cheng Zhang. <i>Proc. ACM Interact. Mob. Wearable</i>                                                                   | Nov. 2024                     |

*Ubiquitous Technol. (IMWUT '24)* 8, 4, Article 189, 30 pages.

- [EchoGuide: Active Acoustic Guidance for LLM-Based Eating Event Analysis from Egocentric Videos.](#) Vineet Parikh, Saif Mahmud, Devansh Agarwal, **Ke Li**, François Guimbretière, and Cheng Zhang. In *Proceedings of the 2024 ACM International Symposium on Wearable Computers (ISWC '24)*, 40–47. **Best Paper Honorable Mention Award** Melbourne, Australia Oct. 2024
- [AI-Powered Eyewear for Routine Facial Expression Analysis in Parkinson's Disease: Study Design and Goals.](#) Hwai Yin Ooi, Cheng Zhang, **Ke Li**, Jordan Narins, and Harini Sarva. *The International Congress of Parkinson's Disease and Movement Disorders*, 39 (suppl 1). Sept. 2024
- [GazeTrak: Exploring Acoustic-based Eye Tracking on a Glass Frame.](#) **Ke Li**, Ruidong Zhang, Boao Chen, Siyuan Chen, Sicheng Yin, Saif Mahmud, Qikang Liang, François Guimbretière, and Cheng Zhang. In *Proceedings of the 30th Annual International Conference on Mobile Computing and Networking (ACM MobiCom '24)*, 497–512. D.C., USA Nov. 2024
- [EyeEcho: Continuous and Low-power Facial Expression Tracking on Glasses.](#) **Ke Li**, Ruidong Zhang, Siyuan Chen, Boao Chen, Mose Sakashita, François Guimbretière, and Cheng Zhang. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*, Article 319, 1–24. Honolulu, USA May 2024
- [EchoWrist: Continuous Hand Pose Tracking and Hand-Object Interaction Recognition Using Low-Power Active Acoustic Sensing On a Wristband.](#) Chi-Jung Lee, Ruidong Zhang, Devansh Agarwal, Tianhong Catherine Yu, Vipin Gunda, Oliver Lopez, James Kim, Sicheng Yin, Boao Dong, **Ke Li**, Mose Sakashita, François Guimbretière, and Cheng Zhang. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*, Article 403, 1–21. Honolulu, USA May 2024
- [HPSpeech: Silent Speech Interface for Commodity Headphones.](#) Ruidong Zhang, Hao Chen, Devansh Agarwal, Richard Jin, **Ke Li**, François Guimbretière, and Cheng Zhang. In *Proceedings of the 2023 ACM International Symposium on Wearable Computers (ISWC '23)*, 60–65. **Best Paper Honorable Mention Award** Cancún, Mexico Oct. 2023
- [EchoNose: Sensing Mouth, Breathing and Tongue Gestures inside Oral Cavity using a Non-contact Nose Interface.](#) Rujia Sun, Xiaohe Zhou, Benjamin Steeper, Ruidong Zhang, Sicheng Yin, **Ke Li**, Shengzhang Wu, Sam Tilsen, François Guimbretière, and Cheng Zhang. In *Proceedings of the 2023 ACM International Symposium on Wearable Computers (ISWC '23)*, 22–26. Cancún, Mexico Oct. 2023
- [PoseSonic: 3D Upper Body Pose Estimation Through Egocentric Acoustic Sensing on Smartglasses.](#) Saif Mahmud, **Ke Li**, Guilin Hu, Hao Chen, Richard Jin, Ruidong Zhang, François Guimbretière, and Cheng Zhang. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. (IMWUT '23)* 7, 3, Article 111, 28 pages. Sept. 2023
- [EchoSpeech: Continuous Silent Speech Recognition on Minimally-obtrusive Eyewear Powered by Acoustic Sensing.](#) Ruidong Zhang, **Ke Li**, Yihong Hao, Yufan Wang, Zhengnan Lai, François Guimbretière, and Cheng Zhang. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*, Article 852, 1–18. Hamburg, Germany Apr. 2023
- [EarIO: A Low-power Acoustic Sensing Earable for Continuously Tracking Detailed Facial Movements.](#) **Ke Li**, Ruidong Zhang, Bo Liang, François Guimbretière, and Cheng Zhang. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. (IMWUT '22)* 6, 2, Article 62, 24 pages. Jul. 2022
- [Locating Everyday Objects using NFC Textiles.](#) Jingxian Wang, Junbo Zhang, **Ke Li**, Chengfeng Pan, Carmel Majidi, and Swarun Kumar. In *Proceedings of the 20th International Conference on Information Processing in Sensor Networks (IPSN '21)*, 15–30. **Best Paper Award & Best** Nashville, USA May 2021

### **Presentation Award**

- [A Constant Factor Approximation for  \$d\$ -Hop Connected Dominating Set in 3-Dimensional Wireless Network.](#) **Ke Li**, Xiaofeng Gao, Fan Wu, and Guihai Chen. In *IEEE Transactions on Wireless Communications (TWC)*, vol. 18, no. 9, pp. 4357-4367, Sept. 2019. Sept. 2019
- [QLEC: A Machine-Learning-Based Energy-Efficient Clustering Algorithm to Prolong Network Lifespan for IoT in High-Dimensional Space.](#) **Ke Li**, Haowei Huang, Xiaofeng Gao, Fan Wu, and Guihai Chen. In *Proceedings of the 48th International Conference on Parallel Processing (ICPP '19)*, Article 105, 1–10. Kyoto, Japan Aug. 2019

### **PATENTS**

- [US20240212388A1: Wearable Devices to Determine Facial Outputs using Acoustic Sensing.](#) **Ke Li**, Cheng Zhang, Francois Guimbretiere, Ruidong Zhang.
- [US12400473B2: Wearable Facial Movement Tracking Devices.](#) Cheng Zhang, Tuochao Chen, Benjamin Steeper, Francois Guimbretiere, **Ke Li**, Ruidong Zhang, Hyunchul Lim.

### **HONORS & AWARDS**

- Qualcomm Innovation Fellowship 2024** Apr. 2024  
*Fellowship focused on recognizing, rewarding, and mentoring PhD students based on Qualcomm's core values of innovation, execution, and teamwork*
- Nominated for a 2021-2022 Outstanding Teaching Award** May 2022  
*Honor bestowed on outstanding PhD teaching assistants in Information Science at Cornell*
- Outstanding Graduate in Shanghai (Top 5%)** Jun. 2020  
*Honor bestowed on outstanding graduates from universities in Shanghai*
- A-level Outstanding Scholarship of Shanghai Jiao Tong University (Top 1%)** Nov. 2019  
*Scholarship awarded to outstanding students at SJTU*
- Tang Lixin Scholarship (Top 0.2%)** Oct. 2019  
*Scholarship founded by Mr. Tang Lixin and awarded to Top 60 students at SJTU*
- National Scholarship\*2 (3/158, 1/158)** Oct. 2018, 2019  
*Top scholarship awarded to undergraduates in China for their achievements in academics*
- B-level Outstanding Scholarship of Shanghai Jiao Tong University\*2 (Top 5%, Top 2%)** Nov. 2017, 2018  
*Scholarship awarded to outstanding students at SJTU*
- Wen-Yuan Pan Scholarship (Top 5%)** Dec. 2017  
*Scholarship founded by Wen-Yuan Pan Foundation*

### **SERVICE**

- External Reviewer** 2022 - Present  
*UbiComp '22 '23 '24 '25 '26, UIST '23 '26, CHI '24 '25 '26*

### **TEACHING EXPERIENCES**

- INFO 4100/5101: Learning Analytics** Spring '26  
*Teaching Assistant at Cornell University, Instructor: Prof. Rene Kizilcec*
- INFO 2951: Introduction to Data Science with R** Spring '25  
*Teaching Assistant at Cornell University, Instructor: Prof. Benjamin Soltoff*
- INFO 4320/5321: Introduction to Rapid Prototyping and Physical Computing** Fall '22 '23, Spring '23

*Teaching Assistant at Cornell University, Instructor: Prof. Cheng Zhang*

**INFO 4120/6120: Ubiquitous Computing** (*Nominated for Outstanding Teaching Award*)

Fall '21

*Teaching Assistant at Cornell University, Instructor: Prof. Cheng Zhang*

## **EXTRACURRICULAR ACTIVITIES & LEADERSHIP**

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**Development Program for Excellent Student of SJTU, Group Leader**

Apr. 2017 – Apr. 2018

- Organized various activities, such as lectures, speech contests, volunteer programs, and summer social activities.
- Awarded Outstanding Participant Title in the 2018 Summer Social Practice of SJTU (Top 2%).

**Master Distinguished Lecture, SJTU, Director**

May. 2017 - Sept. 2018

- Organized professional academic lectures and invited scholars for speeches, including 8 Nobel Prize Laureates.
- Oversaw the entire process of lecture, including brand building, publicity, directing, reception and recognition.